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C:\Users\minfa\AI Test\Write-ups\TokenAwakening>python eval_prompts.py
Extracting text from
C:\Users\minfa\Downloads\2025_Federal_Reserve_Stress_Test_Results.pdf...
Extracting text from
C:\Users\minfa\Downloads\2026_Federal_Reserve_Stress_Test_Results.pdf...
Extraction complete. Read 334338 characters.
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🏃 RUNNING MEDIUM PROMPT | Model: gemini-2.5-flash
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[Output]
Here is the analysis of the 2025 and 2026 CCAR results for Bank of America and Citigroup:

****Calculated Loan Loss Rate Changes and Differences****

Metric	Value (p.p.)
:-----	:-----
BofA CRE YoY Change	0.2
BofA CC YoY Change	-0.3
Citi CRE YoY Change	2.6
Citi CC YoY Change	-0.3
2025 CRE Difference (Citi - BofA)	-0.8
2025 CC Difference (Citi - BofA)	0.1
2026 CRE Difference (Citi - BofA)	1.6
2026 CC Difference (Citi - BofA)	0.1

****Summary of Citigroup's Loss Rates Compared to Bank of America****

In 2025, Citigroup's projected domestic Commercial Real Estate (CRE) loss rate was 0.8 percentage points lower than Bank of America's. However, by 2026, Citigroup's CRE loss rate increased significantly, becoming 1.6 percentage points higher than Bank of America's. For Credit Cards, Citigroup's projected loss rate was consistently slightly higher than Bank of America's in both years, with a difference of 0.1 percentage points in both 2025 and 2026. Overall, Citigroup experienced a substantial relative increase in its CRE loss rate compared to Bank of America from 2025 to 2026, while its Credit Card loss rate maintained a similar slight premium over Bank of America.

****Fed's Qualitative Commentary on Increased Aggregate Loan Losses (2025 to 2026)****

The Federal Reserve's qualitative commentary on why aggregate loan losses increased from 2025 to 2026 is provided in the ****2026 Federal Reserve Stress Test Results**** document, specifically in the ****"Introduction"** section, under **"Summary of Results"** on page 3******.

The document states:

"Projected capital decreased due to greater loan losses as a result of an increase in loan balances and the severity of certain scenario variables: In 2025, loan balances grew by around 10 percent, concentrated in credit

cards and wholesale loans. Combined with more severe paths for certain scenario variables in this year's stress test, including more severe changes in corporate bond spreads and commercial real estate prices, this led to greater loan losses year-over-year."

[Usage] Prompt Tokens: 100,925 | Output Tokens: 2,387
[Cost] Input Cost: \$0.030277 | Output Cost: \$0.005967 | Total Cost: \$0.036245
→ Math passed. Running DeepEval Faithfulness Check...
→ DeepEval Score: 100.0% | Reason: The score is 1.00 because there are no contradictions between the actual output and the retrieval context. Great job—the output stays fully faithful to the provided information!
→ DeepEval Usage: Input 144,833 / Output 5,885 | Total Cost: \$0.871290
✅ Validation: PASSED

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🔥 RUNNING MEDIUM PROMPT | Model: gemini-2.5-pro

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[Output]
Based on my analysis of the 2025 and 2026 CCAR results for Bank of America and Citigroup, please find the requested calculations and summaries below.

1 & 2. Calculated Loan Loss Rate Changes and Differences

The following table presents the year-over-year change in projected loan loss rates for each bank and the difference in loss rates between the two firms for the specified portfolios.

Metric	Value (p.p.)
BofA CRE YoY Change	0.2
BofA CC YoY Change	-0.3
Citi CRE YoY Change	2.6
Citi CC YoY Change	-0.3
2025 CRE Difference (Citi - BofA)	-0.8
2025 CC Difference (Citi - BofA)	0.1
2026 CRE Difference (Citi - BofA)	1.6
2026 CC Difference (Citi - BofA)	0.1

3. Comparison of Citigroup vs. Bank of America Loss Rates

Overall, Citigroup's projected credit card loss rates were marginally higher than Bank of America's by 0.1 percentage points in both the 2025 and 2026 stress tests.

For the domestic commercial real estate portfolio, Citigroup's projected loss rate was 0.8 percentage points lower than Bank of America's in the 2025 test. However, this trend reversed in the 2026 test, where Citigroup's projected CRE loss rate was significantly higher than Bank of America's by 1.6 percentage points, driven by a substantial 2.6

percentage point year-over-year increase in Citigroup's own projected CRE losses.

4. Federal Reserve Commentary on Increased Aggregate Loan Losses

According to the **"2026 Federal Reserve Stress Test Results"** document, the Federal Reserve attributed the year-over-year increase in aggregate loan losses to two primary factors. As stated in the **"Summary of Results"** section on page 3, projected capital decreased due to:

- An increase in loan balances:** The report notes that in 2025, aggregate loan balances grew by approximately 10 percent, with concentration in credit cards and wholesale loans.
- The severity of certain scenario variables:** The 2026 severely adverse scenario featured more severe paths for specific variables, including "more severe changes in... commercial real estate prices," which contributed to greater projected loan losses compared to the prior year.

[Usage] Prompt Tokens: 100,925 | Output Tokens: 2,460
[Cost] Input Cost: \$0.126156 | Output Cost: \$0.024600 | Total Cost: \$0.150756

- Math passed. Running DeepEval Faithfulness Check...
- DeepEval Score: 100.0% | Reason: The score is 1.00 because the actual output is fully faithful to the retrieval context with no contradictions found. Great job maintaining accuracy and alignment!
- DeepEval Usage: Input 145,101 / Output 6,248 | Total Cost: \$0.881705

✅ Validation: PASSED

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🏃 RUNNING MEDIUM PROMPT | Model: claude-sonnet-4-6

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[Output]
Analysis: Bank of America vs. Citigroup Loan Loss Rates (2025 & 2026 CCAR)

Step 1 & 2: Numeric Results

****Source data extracted from the documents:****

Bank	Portfolio	2025 Loss Rate	2026 Loss Rate
Bank of America	CRE, domestic	9.1%	9.3%
Bank of America	Credit cards	16.5%	16.2%
Citigroup	CRE, domestic	8.3%	10.9%
Citigroup	Credit cards	16.6%	16.3%

(Sources: 2025 results – Table 10, p. 22; 2026 results – Table 10, pp. 21-22)

****Calculations:****

- BofA CRE YoY: $9.3 - 9.1 = **+0.2 \text{ p.p.**}$
- BofA CC YoY: $16.2 - 16.5 = **-0.3 \text{ p.p.**}$
- Citi CRE YoY: $10.9 - 8.3 = **+2.6 \text{ p.p.**}$
- Citi CC YoY: $16.3 - 16.6 = **-0.3 \text{ p.p.**}$
- 2025 CRE Difference (Citi - BofA): $8.3 - 9.1 = **-0.8 \text{ p.p.**}$
- 2025 CC Difference (Citi - BofA): $16.6 - 16.5 = **+0.1 \text{ p.p.**}$
- 2026 CRE Difference (Citi - BofA): $10.9 - 9.3 = **+1.6 \text{ p.p.**}$
- 2026 CC Difference (Citi - BofA): $16.3 - 16.2 = **+0.1 \text{ p.p.**}$

Markdown Table

Metric	Value (p.p.)
BofA CRE YoY Change	+0.2
BofA CC YoY Change	-0.3
Citi CRE YoY Change	+2.6
Citi CC YoY Change	-0.3
2025 CRE Difference (Citi - BofA)	-0.8
2025 CC Difference (Citi - BofA)	+0.1
2026 CRE Difference (Citi - BofA)	+1.6
2026 CC Difference (Citi - BofA)	+0.1

Step 3: Qualitative Comparison – Citigroup vs. Bank of America

Commercial Real Estate (CRE):

In 2025, Citigroup's CRE loss rate (8.3%) was actually *lower* than Bank of America's (9.1%) by 0.8 p.p. However, by 2026, the relationship reversed sharply: Citigroup's CRE rate surged to 10.9% – a 2.6 p.p. increase – while Bank of America's rose only modestly to 9.3%. As a result, Citigroup's CRE loss rate exceeded Bank of America's by 1.6 p.p. in 2026. This suggests Citigroup's CRE portfolio carries meaningfully greater sensitivity to the more severe CRE price stress embedded in the 2026 scenario.

Credit Cards:

Across both years, Citigroup's and Bank of America's credit card loss rates were nearly identical. In 2025, Citigroup (16.6%) was just 0.1 p.p. above Bank of America (16.5%), and in 2026, Citigroup (16.3%) remained just 0.1 p.p. above Bank of America (16.2%). Both banks saw a marginal decline of 0.3 p.p. year-over-year. The credit card portfolios of the two institutions thus exhibit very similar stressed loss characteristics, with Citigroup holding a negligible and consistent premium.

****Overall:**** Citigroup's loss profile relative to Bank of America is essentially equivalent on credit cards but diverges materially on CRE, with Citigroup shifting from a slight advantage in 2025 to a notable disadvantage in 2026.

Step 4: Fed's Qualitative Commentary on Drivers of Increased Aggregate Loan Losses (2025 → 2026)

The Federal Reserve's explanation for the increase in aggregate loan losses from the 2025 to the 2026 stress test is found in the **2026 Federal Reserve Stress Test Results, Introduction section, "Summary of Results,"** under the sub-heading discussing drivers of year-over-year changes (pp. 3-4), specifically in the first bullet point of the decomposition discussion and in **Table 2 (Key variables in 2025 and 2026 supervisory severely adverse scenarios, p. 4)**.

The Fed states:

> **"Projected capital decreased due to greater loan losses as a result of an increase in loan balances and the severity of certain scenario variables: In 2025, loan balances grew by around 10 percent, concentrated in credit cards and wholesale loans. Combined with more severe paths for certain scenario variables in this year's stress test, including more severe changes in corporate bond spreads and commercial real estate prices, this led to greater loan losses year-over-year. The scenario changes are due to changes in the U.S. economy over 2025 and the countercyclical design of the Board's annual hypothetical scenario."**

Specifically, the 2026 scenario featured a more severe CRE price shock (-39% in 2026 vs. -30% in 2025, per Table 2, p. 4), directly driving higher CRE loss rates. For credit cards, the primary driver was the approximately 10% growth in credit card loan balances during 2025, which increased the absolute dollar exposure subject to stress, even though the credit card loss rate itself was relatively stable (17.1% aggregate in 2026 vs. 16.9% in 2025 per Tables 9 in each report). The combination of higher balances and a slightly higher aggregate rate produced materially larger credit card loss dollars (\$203 billion in 2026 vs. \$157.5 billion in 2025).

[Usage] Prompt Tokens: 101,516 | Output Tokens: 1,599

[Cost] Input Cost: \$0.304548 | Output Cost: \$0.023985 | Total Cost: \$0.328533

→ Math passed. Running DeepEval Faithfulness Check...

→ DeepEval Score: 100.0% | Reason: The score is 1.00 because the actual output is fully faithful to the retrieval context with no contradictions detected. Great job maintaining accuracy and alignment throughout!

→ DeepEval Usage: Input 146,711 / Output 7,168 | Total Cost: \$0.912755

✅ Validation: PASSED